Project Ouroboros – Hardware Components & Cost Breakdown

This document provides a detailed list of the hardware components required for **Project Ouroboros**, covering both **defensive** (Kickstarter consumer device) and **offensive/research** (open-source) sides. Prices are estimated in **INR** and **USD**, based on common suppliers (AliExpress, Amazon, local Indian electronics markets).

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1. Core Components

Component	Description	Qty	Price (INR)	Price (USD)
ESP32-WROOM/ESP32-S3 Dev Board	Main MCU, Wi-Fi + BLE support, low-cost, TinyML capable	1	₹350–₹500	$4–6
ESP32 with External Antenna (optional)	For extended range in spectrum analysis & jamming tests	1	₹600–₹900	$7–10
OLED 1.3” / TFT 2.4” Display	Compact display for real-time graphs, menus	1	₹350–₹700	$4–8
Touchscreen TFT (2.8–3.2”)	For advanced UI in premium defensive version	1	₹700–₹1200	$8–14
MicroSD Card Module	For logging packets, replay attack profiles	1	₹80–₹120	$1–1.5
Li-Po Battery (1200–2500mAh)	Rechargeable power source	1	₹350–₹600	$4–7
TP4056 Charging Module (with protection)	Li-Po charging & safety	1	₹50–₹100	$0.5–1
Boost Converter 5V	Stable power supply	1	₹60–₹120	$0.7–1.5
Buttons/Rotary Encoder	For manual control if touchscreen not used	2–3	₹50–₹150	$1–2
3D Printed / Acrylic Enclosure	Protective casing, branding-ready	1	₹500–₹1000	$6–12

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2. RF & Wireless Modules

Component	Description	Qty	Price (INR)	Price (USD)
nRF24L01+ Module (with PA+LNA)	For RF sniffing, Zigbee-like 2.4GHz comms	1	₹150–₹300	$2–3
CC1101 Sub-GHz Module	315/433/868/915MHz RF sniffing & replay	1	₹300–₹450	$4–5
HackRF One SDR (optional, advanced)	Full SDR platform (TX/RX, 1–6GHz)	1	₹8,000–₹12,000	$95–140
RTL-SDR USB Dongle (budget option)	RX-only SDR for spectrum analysis	1	₹2,000–₹3,000	$25–35
433MHz Antennas (SMA)	For Sub-GHz experiments	1–2	₹200–₹400	$2–4
2.4GHz Antennas (Wi-Fi/ Bluetooth)	External high-gain antennas	1–2	₹250–₹600	$3–6

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3. Defensive Device-Specific (Kickstarter Version)

Component	Description	Qty	Price (INR)	Price (USD)
ESP32-S3 Dev Board	More powerful, for TinyML + defense monitoring	1	₹600–₹900	$7–10
TFT Display (2.8”–3.2”, Touch)	Full UI with spectrum analyzer	1	₹800–₹1200	$9–14
High-Capacity Li-Po (2500–4000mAh)	Longer runtime for consumer use	1	₹800–₹1200	$9–14
Compact Plastic Case (Injection Mold Ready)	Kickstarter-ready enclosure	1	₹1000–₹1500	$12–18

Estimated Defensive Device Cost (Kickstarter-ready): - **Prototype (DIY build):** ~₹2,500–₹4,000 ($30–48) - **Mass Production (1000+ units, molded case):** ~₹1,200–₹1,800 ($15–22)

Retail Price Target: **₹3,500–₹5,000 ($40–60)**

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4. Offensive Device-Specific (Open Source)

Component	Description	Qty	Price (INR)	Price (USD)
CC1101 + nRF24L01+ Combo	Multi-band RF attacks	1 each	₹450–₹700	$6–8
HackRF One / RTL-SDR	For advanced SDR attacks	1	₹2,000–₹12,000	$25–140
SD Card (16–32GB)	For packet capture/replay storage	1	₹250–₹400	$3–5

Estimated Offensive Toolkit Cost (ESP32 + modules, excluding SDR): - ~₹2,000–₹3,000 ($25–35)

With HackRF: ~₹10,000+ ($120+)

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5. Total Estimated Build Cost

- **Minimal Defensive Shield (ESP32 + OLED + battery + case):** ₹1,200–₹1,800 ($15–22)
- **Advanced Defensive Device (TFT, logging, TinyML, enclosure):** ₹2,500–₹4,000 ($30–48)
- **Full Offensive Toolkit (ESP32 + RF modules, no SDR):** ₹2,000–₹3,000 ($25–35)
- **Offensive Toolkit + SDR (HackRF/RTL-SDR):** ₹10,000–₹15,000 ($120–180)

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6. Notes on Cost Strategy

- **Kickstarter Defensive Version** → Production cost must be kept **below ₹1,800 ($22)** per unit to allow for healthy retail margin.
- **Offensive Open-Source Version** → Users build themselves; cost not a concern, since it won't be sold.
- **Bundle Strategy** → Option to sell **Defensive Shield + optional add-on kit** (antennas, modules, SDR guides) for hobbyists.

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7. Conclusion

- **Project Ouroboros Hardware** can be built very cheaply, starting from **₹1,200 ($15)** for basic defense, scaling to **₹15,000 ($180)** for a full offensive + SDR lab.
- The **dual-strategy** ensures profitability from Kickstarter while ensuring recognition in the hacker/research community via open-source releases.